

Monica Alexander

monicaalexander.com
monica.alexander@utoronto.ca

Academic Appointments

University of Toronto

Associate Professor (with tenure), Statistical Sciences and Sociology July 2024 – *current*
Assistant Professor, Statistical Sciences and Sociology July 2018 – June 2024
(Maternity leave: April 2019 – December 2019, June 2021 – December 2021)

Education

Ph.D (Demography), **University of California, Berkeley**, 2018.

Dissertation: Bayesian Methods for Mortality Estimation.

Committee: Josh Goldstein (chair); Ken Wachter; Dennis Feehan; Jas Sekhon.

M.Arts (Statistics), **University of California, Berkeley**, 2015.

M.Social Research (Demography), **Australian National University**, 2013.

B.Science (First Class Honors) (Maths and Chemistry), **University of Tasmania**, 2008.

Professional Appointments

Member, Technical Advisory Group

UN Interagency Group for Child Mortality Estimation

Affiliate Member

Leverhulme Centre for Demographic Science, University of Oxford

Member, Advisory Committee on Demographic Statistics and Studies

Statistics Canada

Papers

(*) denotes student/trainee author

(+) denotes equal contribution

Peer-reviewed articles

Omenti, R.(*), **Alexander, M.** and Barban, N. ‘Bayesian Indirect Estimation of Historical Fertility in Europe and US Using Online Genealogical Data’. Accepted, *Sociological Methods and Research*.

Schlüter, B. S.(*), Root, L. and **Alexander, M.** ‘Trends and Patterns in Childlessness by Age, Race, Ethnicity, and Education in the United States’. Accepted, *Demography*.

Pejchinovska, M.(*), and **Alexander, M.** ‘A Bayesian Framework to Account for Misclassification Error and Uncertainty in the Estimation of Abortion Incidence. *Studies in Family Planning*. 2026. 57(2): 234-255.

Badolato, L., Decter-Frain, A., Irons, N.J., Miranda, M.L., Walk, E., Zhalieva, E., **Alexander, M.**, Basellini, U. and Zagheni, E. ‘The Limits of Predicting Individual-Level Longevity: Insights From the US Health and Retirement Study’. *Demography*. 2026. 63(1): 351-374.

Ji, Y.(*), **Alexander, M.**, and Craiu, R.V. ‘Contributed Discussion of Model Uncertainty and Missing Data: An Objective Bayesian Perspective’. *Bayesian Analysis*. 2025. 20(4): 1732-1735.

Potter, A. L., Schluter, B. S.(*), **Alexander, M.**, Yang, C. F. J., & Kiang, M. V. ‘Youths Experiencing Parental Death Due to Cancer’. *JAMA Network Open*, 2025. 8(7): e2519106-e2519106.

Cresswell, J.A., **Alexander, M.**, Chong, M.Y.C.(*), Link, H.M., Pejchinovska, M.(*), Gazeley, U., Ahmed, S.M., Chou, D., Moller, A.B., Simpson, D. and Alkema, L., et al. ‘Global and regional causes of maternal deaths 2009–20: a WHO systematic analysis.’ *The Lancet Global Health*, 2025. 13(4): e626–e634.

Dharamshi, A.(*), **Alexander, M.**, Winant, C. and Barbieri, M. ‘Jointly Estimating Sub-national Mortality for Multiple Populations’. *Demographic Research*, 2025. 52(3): 71–110.

Chong, M. Y.C.(*), Pejchinovska, M.(*), and **Alexander, M.**, ‘Estimating causes of maternal death in data-sparse contexts’. *Statistics in Medicine*, 2024. 43(24): 4702–4735.

Alexander, M. and Raftery, A.E., ‘Developing and implementing the UN’s probabilistic population projections as a milestone for Bayesian demography: An interview with Adrian Raftery.’ *Demographic Research*, 2024. 51, pp.1-16.

Kiang M.V. and **Alexander, M.** ‘Invited commentary: Motivating better methods — and better data collection — for measuring prevalence of drug misuse’, *American Journal of Epidemiology*. 2024. 194(1): 12–16.

Schlüter, B. S.(*), Alburez-Gutierrez, D., Bibbins-Domingo, K., **Alexander, M.**(+), and Kiang, M. V.(+) ‘Youth experiencing parental death due to drug poisoning and firearm violence in the US, 1999-2020’. *Journal of the American Medical Association (JAMA)*, 2024. 331(20):1741–1747.

Chong, M.(*), and **Alexander, M.** ‘Estimating the timing of stillbirths in countries worldwide using a Bayesian hierarchical penalized splines regression model.’ *Journal of the Royal Statistical Society Series C*, 2024. 73(4): 902–920.

Katz, L.(*), Chong, M.(*), and **Alexander, M.**, ‘Measuring short-term mobility patterns in North America using Facebook Advertising data, with an application to adjusting Covid-19 mortality rates’. *Demographic Research*, 2024, 50(10): 291–324.

Yeung, J.(*), **Alexander, M.**, and Riffe, T. ‘Bayesian implementation of Rogers-Castro model migration schedules: An alternative technique for parameter estimation’. *Demographic Research*, 2023, 49(42): 1201-1228.

Alexander, M., and Alkema, L., ‘A Bayesian Cohort Component Projection Model to Estimate Women of Reproductive Age at the Subnational Level in Data-Sparse Settings’, *Demography*, 2022. 59(5): 1713–1737.

Maheu-Giroux, M, Sardinha, L, Stöckl, H, Meyer, S, Godin, A, **Alexander, M.**, and García-Moreno, C. ‘A framework to model global, regional, and national estimates of intimate partner violence.’ *BMC medical research methodology*, 2022. 22(1): 1–17.

Susmann, H., **Alexander, M.**, and Alkema, L.. ‘Temporal models for demographic and global health outcomes in multiple populations: Introducing a new framework to review and standardize documentation of model assumptions and facilitate model comparison’. *International Statistical Review*, 2022. 90(3):437-467.

Alexander, M.(+), and Root, L.(+), ‘Competing effects on the average age of infant death’. *Demography*, 2022. 59 (2): 587–605.

Kiang MV, Tsai AC, **Alexander, M.**, Rehkopf DH, Basu S. ‘Racial/Ethnic Disparities in Opioid-Related Mortality in the USA, 1999–2019: the Extreme Case of Washington DC.’ *Journal of Urban Health*, 2021. 98, 589–595.

Wilson, T., Grossman, I., **Alexander, M.**, Rees, P., and Temple, J. ‘Methods for small area population forecasts: state-of-the-art and research needs’, *Population Research and Policy Review*, 2021. 41, 865–898

Kiang MV, Chen JT, Krieger N, Buckee CO, **Alexander, M.**, Baker JT, Buckner RL, Coombs III G, Rich-Edwards JW, Carlson KW, and Onnela J-P. ‘Sociodemographic characteristics of missing data in digital phenotyping’. *Scientific Reports*. 2021. 11(1): 15408.

Alexander, M., Polimis, K., and Zagheni, E., ‘Combining social media and survey data to nowcast migrant stocks in the United States’, *Population Research and Policy Review*, 2020. 41, 1–28.

Alexander, M., Polimis, K., and Zagheni, E., ‘The impact of Hurricane Maria on out-migration from Puerto Rico: Evidence from Facebook data’, *Population and Development Review*, 2019, 45(3): 617-630.

Hug, L., **Alexander, M.**, You, D., and Alkema, L. ‘National, regional, and global levels and trends in neonatal mortality between 1990 and 2017, with scenario-based projections to 2030: a systematic analysis by the United Nations Inter-agency Group for Child Mortality Estimation’, *Lancet Global Health*, 2019, 7(6): 710–720.

Kiang, M.V., Basu S., Chen J., and **Alexander, M.** ‘Assessment of Changes in the Geographical Distribution of Opioid-Related Mortality Across the United States by Opioid Type, 1999-2016’. *JAMA Network Open*. 2019, 2(2):e190040.

Gemmill, A., Kiang, M.V., and **Alexander, M.** ‘Trends in pregnancy-associated mortality involving opioids in the United States, 2007–2016’, *American Journal of Obstetrics and Gynecology*, 2019, 220(1): 115–116.

Alexander, M.(+), Kiang, M.V.(+), and Barbieri M., ‘Trends in Black and White Opioid Mortality in the United States, 1979–2015’, *Epidemiology*, 2018, 29(5): 707–715.

Alexander, M., and Alkema, L., ‘Global Estimation of Neonatal Mortality using a Bayesian Hierarchical Splines Regression Model’, *Demographic Research*, 2018, 38(15): 335–372.

Alexander, M., Zagheni, E., and Barbieri, M., ‘A Flexible Bayesian Model for Estimating Subnational Mortality’, *Demography*, 2017, 54(6): 2025–2041.

Howlett, M., Gray, M., and Hunter, B., ‘Wages, government payments and other income of Indigenous and non-Indigenous Australians’, *Australian Journal of Labour Economics*, 2016, 19(2): 53–76.

Hunter, B., **Howlett, M.**, and Gray, M., ‘The Economic Impact of the Mining Boom on Indigenous and Non-Indigenous Australians’, *Asia & the Pacific Policy Studies*, 2015, 2(3): 517–530.

Gray, M., **Howlett, M.**, and Hunter, B., ‘Labour Market Outcomes for Indigenous Australians’, *The Economic and Labour Relations Review*, 2014, 25(3): 497–517.

Biddle, N., **Howlett, M.**, Hunter, B., and Paradies, Y., ‘Labour Market and Other Discrimination Facing Indigenous Australians’, *Australian Journal of Labour Economics*, 2013, 16(1): 91–113.

Book chapters

Alexander, M. ‘Using social media advertising data to estimate migration trends over time’. In *Big Data Applications in Geography and Planning*. Edward Elgar Publishing. 2021.

Other published papers

Chong, M.(*), Alburez-Gutierrez, D., Del Fava, E., **Alexander, M.**, & Zagheni, E. 2022. ‘Identifying and correcting bias in big crowd-sourced online genealogies’ (No. WP-2022-005). Max Planck Institute for Demographic Research, Rostock, Germany.

Winant, C., **Alexander, M.**, Dharamshi, A., Dukhovnov, D., & Barbieri, M. 2021. Methods Protocol for the United States Mortality County Database.
<https://usa.mortality.org/uploads/counties/USCountyBayesianEstimationMethodsProtocol20210927.pdf>.

Hunter, B., **Howlett, M.**, and Biddle, N., ‘Modelling Exposure to Risk of Experiencing Discrimination in the Context of Endogenous Ethnic Identification’, *IZA Discussion Paper #8040*, March 2014.

Gray, M., Hunter, B., and **Howlett, M.**, ‘Indigenous Employment: A Story of Continuing Growth’, *CAEPR Topical Issue 2/2013*, Australian National University, Canberra.

Working papers

McNellan, C., Marquez, N. and **Alexander, M.** ‘Homicide in Pregnant and Postpartum versus Nonpregnant and Nonpostpartum Populations: Re-estimation of a Rate Ratio using a Person-time Framework’.
<https://www.medrxiv.org/content/10.64898/2026.01.25.26344756v1>.

Alexander, M. ‘Decomposing dimensions of mortality inequality’.
<https://osf.io/preprints/socarxiv/uqwxj>.

Alexander, M., M. and Root, L. ‘Racial disparities in premature births, fetal deaths and neonatal deaths: a multi-decrement lifetable approach’.
<https://osf.io/preprints/socarxiv/k5qp7/>.

McBrien, H.(*), and **Alexander, M.** ‘Trends in opioid seizure data and their association with opioid mortality’.
<https://www.medrxiv.org/content/10.1101/2022.10.08.22280845v1>.

Alexander, M., and Goldstein, J., ‘Deaths without denominators: using a matched dataset to study mortality patterns in the United States’,
<https://osf.io/preprints/socarxiv/q79ye/>.

Grants awarded

Principal Investigator:

Data Sciences Institute	2025-2026
Data Access Grant	
CAD \$10,000	
Institutional Strategic Initiatives	2025
Stage 1 Grant: Toronto Population Network	
CAD \$70,000	
Social Sciences and Humanities Research Council	2023–2025
Insight Grant	
<i>Deficits of migration in the age of Covid-19</i>	
CAD \$60,000	
Data Sciences Institute, University of Toronto	2023-2024
Catalyst Grant	
<i>Inequality in childcare: The case of nannies in Canada. (co-PI)</i>	
CAD \$100,000	
Data Sciences Institute, University of Toronto	2023-2024
Computational and Quantitative Social Sciences Grant	
<i>Deficits of migration in the age of Covid-19: A new approach to studying changing migration patterns.</i>	
CAD \$10,000	
School of Cities, University of Toronto	2021-2022
Urban Challenge Grant	
<i>Disparities in climate-induced health outcomes in the Greater Toronto Area. (co-PI)</i>	
CAD \$75,000	
Data Sciences Institute	2021-2022
Data Access Grant	
<i>To purchase data for research on disparities in climate-induced health outcomes in the Greater Toronto Area. (co-PI)</i>	
CAD \$10,000	
Social Sciences and Humanities Research Council	2019–2023
Insight Development Grant	
<i>Using Facebook data to measure and understand international migration patterns in Canada</i>	
CAD \$59,000	
University of Toronto Connaught New Researcher Award	2019–2021
CAD \$18,900	

Partner Investigator:

National Science Foundation	2024–2027
<i>The United States Fertility Database: Understanding and Monitoring Changes in Local Fertility Patterns</i>	
Australian Research Council	2019–2022
Discovery Project Grant	
<i>Developing Robust Small Area Population Forecasts for Planning and Policy</i>	

Teaching

University of Toronto

Special Topics in Applied Statistics (graduate, statistics)	Winter, 2026
Demographic Methods (graduate, statistics and sociology)	Winter, 2019, 2026
Applied Statistics II (graduate, statistics)	Winter, 2020–2024
Statistics for sociologists (graduate, sociology)	Winter, 2023
Intermediate data analysis (graduate, sociology)	Winter, 2021, 2022, 2024
Intermediate quantitative methods (undergraduate, sociology)	Fall 2020, Winter 2023

University of California, Berkeley

<i>Instructor</i> , Formal Demography Workshop	June, 2017
<i>Instructor</i> , Formal Demography Workshop	August, 2015
<i>Graduate Student Instructor</i> , Demographic Methods	Fall Semester, 2014

Other

<i>Instructor</i> , <i>Code Horizons</i>	2023–
Designed and taught workshop on non-traditional data sources in demography	April 2022
Materials: https://mjalexander.github.io/demopop-workshop/	
Designed and taught workshop on social media in demographic research	April 2021
Materials: https://mjalexander.github.io/social_media_workshop/	

Conference presentations and invited talks

Listed for the past five years.

2026

Shifting composition of substances in drug-related mortality by race/ethnicity in the United States, 1999-2024

Population Association of America (PAA) Annual Meeting (April)

Fertility Change in the United States: A Subnational Analysis, 1982-2024

Population Association of America (PAA) Annual Meeting (April)

2025

Estimating historical mortality rates using crowd-sourced online genealogies

CARONTE workshop on mortality models (August)

Accounting for dependencies in fertility and mortality

IUSSP International Population Conference (July)

Estimating historical mortality rates using crowd-sourced online genealogies

Biostatistics Seminar, Wake Forest University (May)

A Bayesian parametric model to estimate male fertility

Population Association of America (PAA) Annual Meeting (April)

Measuring and understanding inequalities in parental loss

Sociology Seminar, University of Oxford (March)

Estimating historical mortality rates using crowd-sourced online genealogies

Statistics Seminar, Imperial College (January)

2024

Estimating childlessness by age: a Bayesian parametric approach

Bayesian Methods for the Social Sciences Workshop, University of Amsterdam (October)

Estimating cause of death in data-sparse contexts: recent progress and future challenges

CARONTE workshop on advances in causes of death modelling, University of Padova (July)

Estimating stillbirth rates by timing

International Association of Official Statistics (IAOS) - International Statistical Institute (ISI) Meeting (May)

Balancing different information sources and modeling decisions in the Bayesian estimation of demographic quantities

Population Association of America (PAA) Annual Meeting (April)

2023

Balancing different information sources and modeling decisions in the Bayesian estimation of demographic quantities

British Society of Population Studies (BSPS) Annual Meeting (September)

A Bayesian framework to account for misclassification error and uncertainty in the estimation of abortion incidence

American Sociological Association (ASA) Annual Meeting (August)

Using social media data to estimate migration

Joint Statistical Meetings (JSM) (July)

Estimating stillbirth rates by timing

International Statistical Institute (ISI) World Statistics Congress (July)

Bayesian demography: a brief history, recent applications, and future directions

Statistics Seminar Series, *University of Washington* (April)

Racial disparities in fetal and infant outcomes: a multiple decrement lifetable approach

Population Association of America (PAA) Annual Meeting (April)

Racial disparities in fetal and infant outcomes: insights from, and for, formal demography

Demography Seminar Series, *Australian National University* (June)

Nuffield Sociology Seminar Series, *Oxford University* (March)

2022

Estimating stillbirth rates by timing

Institute of Mathematical Statistics (IMS) International Conference on Statistics and Data Science (December)

Competing Effects on the Average Age of Infant Death: Reconsidering the Approximation of ${}_1a_0$

Social Statistics Seminar, *University of Manchester* (November)

Racial disparities in infant outcomes: insights from, and for, formal demography

Social Demography Seminar, *Harvard University* (November)

Jointly estimating subnational mortality for multiple populations

Bayesian Methods for the Social Sciences Workshop (October)

Stellar Stats Workshop, *University of Toronto* (May)**Future Projections of State-Level Foster Care Populations in the United States**

PAA Annual Meeting (April)

Reproducibility in DemographyToronto Workshop on Reproducibility, *University of Toronto* (February)**Bayesian models and reconciling incomplete demographic data**Demography Seminar, *University of Wisconsin Madison* (February)**2021****The future of data-driven demographic research** (Panel member)

Max Planck Institute for Demographic Research 25th Anniversary (October)

A Bayesian cohort component projection model for estimating subnational populationsDemography Seminar Series, *Australian National University* (May)Centre for Population Change Seminar Series, *University of Southampton* (April)Population Center Brown Bag, *Florida State University* (February)Space Time Applied Bayesian (STAB) Working Group, *University of Washington* (February)**Overcoming barriers to sharing code**Toronto Workshop on Reproducibility, *University of Toronto* (February)**Using social media data to estimate migration**Mathematics and Statistics Seminar, *Maynooth University* (February)**Software**Author, *rcbayes*, R package to estimate and visualize Rogers-Castro migration schedules in Bayesian framework.<https://cran.uib.no/web/packages/rcbayes/>Contributor, *DemoTools*. R package for demographic analysis.<https://timriffe.github.io/DemoTools/>**Honors****IUSSP Early Career Award (North American region)** 2025**Nuffield College Visitorship, University of Oxford** 2025**Max Planck Sabbatical Award** 2021-2023**Doctoral Completion Fellowship** (*UC Berkeley*) 2017-2018**Elizabeth Scott Memorial Award** (*UC Berkeley*) 2015

Statistics MA student showing the most promise in statistical research.

Regents-Intern Fellowship (*UC Berkeley*) 2013-2015**Lado Ruzicka Prize in Social Research** (*Australian National University*) 2013

Awarded to the most outstanding Master of Social Research candidate.

- W.D. Borrie Prize** (*National essay competition, Masters category*) 2012
For best student paper on a population-related topic.
- University Medal** (*University of Tasmania*) 2008
One of three awarded to the most outstanding Science and Technology undergraduates.
- Edith Rita Lowenstern Prize** (*University of Tasmania*) 2008
Awarded to the most outstanding mathematics honors student.

Previous Professional Experience

- UN Interagency Group on Stateless Populations Estimation**
Lead statistical consultant January 2021 – June 2022
- World Health Organization Group on Estimation of Causes of Maternal Mortality**
Lead statistical consultant January 2019 – December 2020
- World Health Organization**
Consultant September 2016 – June 2017
- Data Science for Social Good**
Fellow May 2016 – September 2016
- Human Mortality Database**
Graduate Student Researcher January 2015 – May 2016
- UNICEF Technical Advisory Group**
Consultant March 2014 – July 2015
- The Centre for Aboriginal Economic Policy Research**
Research Officer April 2012 – December 2014
- Reserve Bank of Australia**
Analyst/Senior Analyst February 2010 – June 2013

Service

Editorial

- Editor:** Special Collection on Formal Relationships, *Demographic Research* (with Victor Guerrero)
- Associate Editor:** *Demographic Research*
- Associate Editor:** *Data Science in Science*
- Editorial Board:** *Demography, Canadian Studies in Population*

University and community

- Co-founder and lead:** Toronto Population Network
- Treasurer:** *Bayesian Social Sciences Section of ISBA.*
- Panel Member:** *International Union of Scientific Study of Populations panel on Digital Demography.*
- Team Member:** *Consortium on Analytics for Data-Driven Decision-Making (CAnD3).*

Co-founder and organizer: Formal Demography Working Group. https://formaldemography.github.io/working_group/

Co-founder: *R-Ladies Toronto*

Women in Demography: Maintain a list of active scholars: https://www.monicaalexander.com/women_scholars/

Professional Memberships

American Sociological Association

American Statistical Association

International Society for Bayesian Analysis

International Statistical Institute

International Union for the Scientific Study of Populations

Institute of Mathematical Statistics

Population Association of America